Project Design Phase-II: Technology Stack

Field	Details
Date	02 November 2026
Team ID	N/A (Individual Project)
Project Name	EV Value & Longevity Analytics Dashboard
Prepared by	Piyush Malviya
Maximum Marks	2 Marks

Define Your Technology Stack

S.No	Component	Description	Technology	Justification / Purpose
1	Query / Calculation Language	Create measures and logic.	Tableau calculated fields and LOD expressions	Build Value Index, averages and percent-of-records measures.
2	Data Storage	Where the data lives.	CSV and CSV.GZ files	Source files for 200,000 training and 100,000 test vehicles.
3	Data Ingestion	Load files into the tool.	Tableau data connection (text file)	Reads both files with identical 37-column schema.
4	Validation	Check quality before charting.	Python 3 with pandas	Independent totals, null counts, duplicate check.
5	Transformation	Clean and shape data.	Tableau Data Source tab	Set data types, rename fields, hide unused columns.
6	Visualization	Build charts and dashboards.	Tableau Desktop (Public edition)	Bars, histogram, map, pie, scatter, treemap, line charts and a dashboard.
7	Presentation / UI	How users view results.	Tableau Public (web)	Interactive filters and story without installing software.